

BREAKHOOK · DESIGN DECISION RECORD

BreakHook TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	BHK-DDR-001	Version	0.1
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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/breakhook		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	TRL 2 review decisions, made under Amish's 2026-10-03 pre-approval

0001: TRL 2 review decisions

- **Date:** 2026-10-03
- **Status:** accepted
- **Decided by:** Amish Chadha, by pre-approval: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." Each recommendation below was made in the TRL 2 review of this repo and is recorded as decided under that pre-approval.

Safety: BreakHook pulls a structure down with a rope near a fire. Every decision that touches safety below takes the conservative option and states what evidence would relax it.

■ Context

The TRL 1 scaffold (BHK-PRC-001 v0.1) left the pole material, the hook construction, the rope system, the hauling crew and five open questions in BHK-PRB-001 undecided. TRL 3 needs each settled so the design can be calculated, modelled and planned.

■ Decisions

#	Decision	Options considered	Why	Safety note and what would relax it
1	Pole: foam-filled, electrical-grade pultruded fibreglass round tube 44.5 x 3.2, three 1.8 m sections, with the maker's ASTM F711 dielectric certificate	Hollow fibreglass tube; bamboo; aluminium	Light, stiff enough, and the only option whose insulation can be certified	Conservative: foam fill and certificate even though the pole is never meant to touch a line. Relax only with a field study showing no wiring hazard, which is unlikely
2	Pole is withdrawn before the haul; the rope alone pulls	Pole stays on and is pulled with the rope; pole used as a lever	Keeps the pole team out of the fall zone and keeps the pole out of the load path	Conservative. No evidence would relax it
3	Hook head: plate cut from 10 mm steel, welded into a slotted steel socket; WLL 3 kN; proof 6 kN before issue	Forged hook; bought pike-pole head	Makeable by any local welder; the pike-pole heads on sale are not rated for a rope pull	Proof load before issue and after any seal break; relax to sample proof loading only after a batch of 20 heads all pass
4	Rope system: 1.5 m galvanised wire rope leader, rated bow shackles, 25 m of 12 mm polyester double braid	Rope tied straight to the hook; chain leader	Leader keeps heat and sheet edges off the rope; chain is too heavy at the pole tip	Factors 6 or more on the rope and leader; relax to a lighter rope only with test data
5	Crew: four to deploy (pole team of two, caller, fall zone marker) and about ten haulers on toggles; requirement R13 added (300 N each)	Four haulers; a pulley or capstan	The pull needs about 3 kN (BHK-CAL-001); a pulley needs a strong anchor that settlements rarely have	Pulley left out; it could be added as an option where a tested anchor (a stake system) exists
6	Hauling distance: nearest hauler at 1.5 x the dwelling height and never closer than 10 m (R3 tightened)	1.5 x height only	Keeps haulers well outside the fall zone and further from heat	Conservative; relax only with fall-zone measurements from TRL 4 trials
7	Overhead lines: never raise the pole within 3 m of any overhead or informal line, whatever the pole's insulation	Rely on the insulated pole	Informal wiring is unpredictable; the hook and wet rope conduct	Conservative; no evidence would relax it
8	Hand band 1.55 m from the butt; hands stay below it; insulated length 3.0 m or more (R14 added)	No marking	Gives the insulated length a meaning in use	None
9	Misuse safeguard: settlement committee holds the kit in a locked, sealed rack with two named keyholders and a use log; pulls only during a fire under the community fire plan	Open storage; fire service storage only	Answers the dispute and eviction concern while keeping the kit near the homes	Conservative
10	Firebreak rule for training: at least two dwellings ahead of the burning one, and only if the pull can finish first; otherwise evacuate	One dwelling ahead	Spread can take under a minute	Relax only with evidence from fire-spread studies in the partner's settlements

#	Decision	Options considered	Why	Safety note and what would relax it
11	Scope: free-standing dwellings and end units only; no shared-wall rows, multi-storey or masonry	Allow shared walls	A shared wall can bring down the neighbour or fall unpredictably	Conservative; relax only after trials on a shared-wall test frame
12	Kit storage and checks: monthly check card; a broken seal means a full check and a fresh proof load	Annual check	Outdoor storage and community use	None
13	Co-design partner, first candidate to approach (not agreed): City of Cape Town Fire and Rescue Service volunteer reservists with a Western Cape settlement committee; second: Kenya Red Cross Society, Nairobi	Others in BHK-PRB-001	Strongest evidence base (Stellenbosch and Imizamo Yethu studies) and an existing reservist system	Not a safety decision
14	Requirements: targets of R1 to R10 kept, R6 measured to hook set with the pole withdrawn and the store within 100 m; R11 to R14 added	Leave as TRL 1	Makes each target measurable	None
15	Budget: <code>budget_usd</code> stays at USD 2,000 as a value-engineering target	Change it	Amish's pre-approval accepts variations; the estimate is under the target anyway	Not a safety decision

■ Consequences

- BHK-REQ-001 v0.2, BHK-PRC-001 v0.2 and BHK-PRB-001 v0.2 carry these decisions.
- The design for construction (BHK-DDR-002) builds on decisions 1 to 4.
- The check card, training material and fall zone kit are part of the community kit (BOM 19 to 24).